

Vikash Kumar Singh

Aspiring Data Scientist | AI Engineer

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PROFILE SUMMARY

AI Engineer and Data Scientist with hands-on experience in LLM pipelines, RAG architectures, and end-to-end ML systems using LangChain, Python, TensorFlow, and Scikit-learn. Skilled in EDA, hypothesis testing, statistical modeling, feature engineering, and deploying scalable AI solutions that translate complex business problems into measurable impact.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain, Ollama, RAG (Retrieval-Augmented Generation), LLM Fine-tuning, Prompt Engineering, Hugging Face, BERT, Transformer, Vector DB (ChromaDB, FAISS), FastAPI

ML & Deep Learning: Scikit-learn, Decision tree, SVM, XGBoost, Linear Regression, PCA TensorFlow, Keras, CNN, ANN, RNN, LSTM, Supervised & Unsupervised Learning, Hyperparameter Tuning (Keras Tuner), Transfer Learning, Streamlit

NLP & Computer Vision: NLP, TF-IDF, Word Embeddings, Object Detection, Image Preprocessing, Data Augmentation

Programming & Data: Python, SQL, NoSQL, Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, Statistical Modeling, Hypothesis Testing

Tools & Platforms: Git, Jupyter Notebook, Google Colab, MySQL, MongoDB, Model Deployment (Basic), Power BI, Tableau

Soft Skills: Cross-functional Collaboration, Stakeholder Management, Project Management, Problem-Solving, Analytical Thinking, Communication & Data Storytelling, Adaptability, Self-driven Learning

PROFESSIONAL EXPERIENCE

AI Engineer / Data Science Consultant

Rubix AI Solutions

Sep 2025 – Mar 2026

Bengaluru, Karnataka, India

- Designed and developed LLM-powered pipelines using LangChain and Ollama, integrating RAG architectures to build context-aware, domain-specific AI applications.
- Built and deployed end-to-end machine learning models (Scikit-learn, TensorFlow) with hyperparameter optimization via Keras Tuner, improving model generalization and accuracy by 15%.
- Performed EDA, feature engineering, and statistical modeling on large datasets to extract actionable insights for business decision-making.
- Collaborated with cross-functional teams to deploy scalable AI/ML solutions; applied data augmentation to boost dataset size by 40% and improve model robustness.

Student Coordinator

Excel Engineering College

May 2024 – Apr 2025

Tamil Nadu, India

- Organized 10+ technical events driving 40% increase in participation; managed and mentored 50+ students through structured planning and stakeholder coordination.

PROJECTS

RAG-Based Document QA System | LangChain, Ollama, RAG, Chroma, Python, Hugging Face

[\[GitHub\]](#)

- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain and Ollama (local LLM) to enable intelligent document question-answering over custom knowledge bases.
- Integrated Chrome vector store with Hugging Face embeddings for semantic retrieval; reduced hallucination by grounding responses in retrieved context.

Multi-Task NLP Classifier | Python, Scikit-learn, NLP, TF-IDF, BERT, Word Embeddings

[\[GitHub\]](#)

- Developed a multi-task NLP model for emotion, hate speech, and violence classification achieving **90%+ accuracy** and reducing inference time by **30%**; evaluated with precision, recall, and F1-score.

Credit Risk Prediction | Python, Pandas, Scikit-learn, ROC-AUC, A/B Testing

[\[GitHub\]](#)

- Built a credit risk ML model achieving 0.87 ROC-AUC on imbalanced datasets; performed EDA, data cleaning, and feature engineering improving performance by **15%**.

Gesture Volume Control | Python, MediaPipe, OpenCV, Pycaw, NumPy

[\[GitHub\]](#)

- Developed a real-time hand gesture-based volume control system using MediaPipe for 21-point hand landmark detection and OpenCV for live webcam feed processing; mapped finger distance to system volume via Pycaw and NumPy interpolation

EDUCATION

Anna University – Excel Engineering College

B.Tech in Computer Science and Business Systems | CGPA: 8.2

- **Honors:** Best Outgoing Student Award

Dec 2021 – May 2025

Tamil Nadu, India

CERTIFICATIONS

- Certified Data Scientist – **NASSCOM** | Certified Data Scientist – **IABAC** | Data Science & ML Certification – **DataMites**